



MOHAMAD ISSA

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Neuroengineer · Researcher · Future Founder · Triathlete — Ironman 70.3

PROFESSIONAL SUMMARY

Neuroengineer with a Master's from TUM (GPA 1.7) and 6+ years spanning brain-computer interfaces, VR neurorehabilitation, myoelectric control, medical software, and clinical application development. IEEE-published, with hands-on patient study experience and a full-stack command of the translational pipeline, from laboratory hypothesis to deployed clinical tool. Currently applying precision software expertise in industry while actively positioning toward founding a neurotechnology company. Driven by the conviction that the most meaningful advances in brain health will come from people who can speak fluently across neuroscience, engineering, and business.

CORE COMPETENCIES

- Neurorehab Technology R&D
- Clinical Application Development
- Brain-Computer Interfaces (BCI/EEG)
- VR/AR Therapeutic Applications
- Experimental Design & User Studies
- Patient-Centered Design & Validation
- Myoelectric Control Systems
- Medical Device Software Development
- Signal Processing & ML Algorithms
- Medical Image Processing (DICOM/NIfTI)
- Scientific Publication & Research
- Translational Development Pipeline
- Agile Development & Project Leadership

PROFESSIONAL EXPERIENCE

Researcher – Master's Thesis: Motor-Cognitive Effects of VR Myoelectric Control

Professorship for Healthcare & Rehabilitation Robotics, TUM | January – August 2023

- Designed and conducted experimental study investigating motor-cognitive effects of VR-based myoelectric training, contributing to fundamental understanding of prosthetic control learning
- Developed novel VR assessment framework combining functional tasks with cognitive workload measures using Unity/C#
- Implemented and compared multiple myoelectric control algorithms (state machine and ML-based) using Python, analyzing performance trade-offs
- Published peer-reviewed research in IEEE RO-MAN 2023, advancing scientific knowledge in rehabilitation robotics

Neuroengineer Research Assistant – Brain-Computer Interfaces

Brain Machine Interface Lab (ICS), TUM | Munich, Germany | October 2021 – January 2023

- Conducted applied research developing EEG-controlled rehabilitation game for quadriplegic patients using Unity and real-time signal processing
- Designed and executed user study with patient, collecting qualitative and quantitative feedback to validate design decisions

- Developed novel AR intervention for Neglect Syndrome, demonstrating translational research from neuroscience principles to clinical tool
- Published article on gamification and motivation in motor rehabilitation, contributing to theoretical understanding of engagement in therapy
- Collaborated with clinical partners to ensure interventions aligned with rehabilitation protocols and patient needs

Lead VR Engineer – Therapy Applications

Humans and Machines Lab, American University of Beirut | Beirut, Lebanon | May 2019 – December 2020

- Led VR development team for therapeutic applications research, managing application development and experimental content creation
- Designed and implemented VR therapy application for Multiple Sclerosis research, contributing to evidence base for VR interventions in neurological rehabilitation
- Coordinated multidisciplinary research team combining clinical, technical, and design perspectives
- Collaborated with clinical partners from the neuroscience institute during the development and to ensure clinical relevance and translational potential of research findings

Medical Software Engineer – Anatomical Mapping

Snke OS | Munich, Germany | January – September 2023

- Developed clinical software tools for processing medical images in DICOM and NIFTI formats for surgical planning applications
- Engineered product features in Python, collaborating with multidisciplinary teams including clinicians and product managers
- Ensured timely delivery of clinical software updates through agile development practices and stakeholder communication

Software Engineer – Cranial Navigation Team

Brainlab | Munich, Germany | March – August 2022

- Developed medical image processing applications in C++ for surgical navigation systems
- Created frontend user interfaces in C# for clinical workflows in operating room environments

Research Intern – Computational Neuroscience

Brain Research Group, Max Planck Institute | Frankfurt, Germany | May – September 2021

- Conducted computational neuroscience research modeling neural assemblies using Python, investigating fundamental mechanisms of brain function

Automation & Software Engineer

CyberTechnologies | Munich, Germany | November 2023 – Present

- Develop and maintain software for semiconductor wafer inspection and metrology systems, ensuring reliability in precision measurement environments

- Lead customer engagement, site installations, and technical troubleshooting, translating complex requirements into practical solutions
- Drive feature development, technical architecture, and quality assurance across full software development lifecycle
- Oversee technical architecture, quality assurance, and debugging of complex software systems, ensuring reliability in production environments
- Apply rigorous problem-solving skills to complex technical challenges, building expertise in precision instrumentation transferable to medical/rehabilitation technologies

EDUCATION

Elite Master of Science in Neuroengineering

Technical University of Munich (TUM) | Munich, Germany | November 2020 – August 2023 | GPA: 1.7/1.0

Bachelor of Engineering in Electrical and Computer Engineering

American University of Beirut (AUB) | Beirut, Lebanon | September 2015 – December 2019 (Minor in Control and Robotics)

SELECTED PUBLICATIONS & RESEARCH

- M. Issa, T. S. Castañeda, P. Capsi-Morales, C. Piazza. "Motor-Cognitive Effects of Virtual Reality Myoelectric Control Training." IEEE RO-MAN 2023.
- M. Issa. "Motivation in Motor Rehabilitation." LinkedIn Article, 2021. Focus on gamification in neurorehabilitation.
- M. Issa, C. Bardawil, N. Daher. "Estimating Braking Torque Uncertainties at the Individual Wheel Level." IEEE IMCET 2018.

TECHNICAL SKILLS

Programming: Python, C++, C#, C, MATLAB

Clinical Technologies: Medical imaging (DICOM, NIfTI), EEG signal processing, EMG/Myoelectric control

Development Platforms: Unity 3D, VR/AR development, TensorFlow, PyTorch

Methodologies: Agile/Scrum, User-centered design, Clinical requirements analysis

Tools: Jira, Git, Simulink, Arduino, MS Office Suite

CERTIFICATIONS & SPECIALIZED TRAINING

- Munich Brain Course – Ludwig-Maximilians-Universität München (2023): Intensive neuroanatomy training with hands-on brain dissection
- Organizer, Neuroengineering Project Week (2 consecutive years) – TUM: Coordinated welcome program for incoming MSNE students
- Automatica 2023 Exhibitor – Presented rehabilitation robotics projects to industry professionals and researchers

LANGUAGES

Arabic (Native) • English (Full Professional Proficiency) • German (B1-B2 Level)